

DOCUMENT 00 · HYBRID HOST ENGINEERING BASELINE

MCRS HYBRID

Modular Captured Ride System · CMFSB
Captured Mechanical Floating Safety Body

Two hybrid (ICE + e-drive) reference hosts · V-L (4-tower Class S) · V-H (Class L) · COMFORT / SPORT captured upper body

Field	Controlled value
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This document is a design baseline and diligence instrument. It does not certify range, crashworthiness, homologation, ASIL, production readiness, patentability or freedom to operate. OPEN items remain OPEN until evidence closes them. Module values are controlled exclusively by Master MCRS-DOC-00-R8.01; this document controls hybrid-host values only.

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1. Purpose, scope and positioning

This baseline defines the integration of the CMFSB isolation module (MCRS-MOD-001) on two hybrid reference hosts (ICE + e-drive): V-L (4×4, four seats) and V-H (6×6, six seats). It is the **hybrid-host controlling Document 00** under the module-centric Master MCRS-DOC-00-R8.01. The Master governs all module values; this document governs hybrid-host values and mirrors only those module sections required for standalone readability.

Document	Role	Governs
MCRS-DOC-00-R8.01	Module-centric master with Annex E and Annex H	Module values; cross-host comparison
MCRS-DOC-00-R5.34 (this file)	Hybrid-host baseline	Hybrid-host values only
MCRS-FE-DOC-00-R7.02 / later	Electric-host baseline	Electric-host values (subordinate to Master for module)

One value, one place. Module sections appearing here are mirrored from the Master by citation; the Master governs. Host values live in §5–§8 and §13. Differences between hybrid and electric hosts are stated explicitly in §2.3. No host powertrain numbers appear in the Master except the comparison table.

1.1 Value statement and product framing

Occupant compartments in specialty, defence-utility, ambulance and limited-series vehicles receive road vibration that degrades comfort, task performance and, where relevant, clinical stability. Soft isolation alone can reduce handling stiffness in rapid manoeuvres. CMFSB addresses both: the upper body stays mechanically captured at all times, with a COMFORT isolation envelope, a SPORT mechanical lock and passive support on power loss. Reference application RA-1 (ambulance / specialty medical transport) is the context for comfort-related requirements; **no clinical outcome is claimed**.

State	Occupant / operator experience	Architecture behaviour	ID
COMFORT	Road harshness isolated from the cabin	Locks released; macro and micro stages active; leveling active	S2
SPORT	Body feels planted for rapid manoeuvres	All required locks confirmed; isolation stages held	S2S
Power loss	Cabin settles on a known mechanical path	Powered valves fail-closed; passive support; no software required	S4

1.2 What is offered

- **Path A (primary):** licence of the module under a paid Option (SIM-001 run and ICD-001 draft); the buyer may bring its own host.
- **Path B:** host engagement studies using the hybrid reference host; no vehicle sale is implied.
- **Path C:** IP-transaction discussion under counsel (T3 only).

2. Executive summary

The CMFSB module places a captured, isolated upper safety body on four or six capture towers. Safety rests on mechanical capture, passive support and stops — never on the actuator or on software. The hybrid hosts supply an ICE + e-drive powertrain, a 48 V / 12 V and small HV electrical system, a fuel system and the mass budget against which the module is evaluated.

Hybrid host item	V-L	V-H	Status
Architecture	Hybrid 4×4, four towers (Class S)	Hybrid 6×6, four towers (Class L); six-tower option under G1	BASELINE
ICE class	V8, 4.0 L twin-turbo class	V12, 6.0 L twin-turbo class	TARGET
Electric traction / battery	≈150–180 kW / 20 kWh usable	≈200–250 kW / 40 kWh usable	TARGET
Fuel tank / range target	80 L / 1,000 km (cycle ≤90–95 km/h avg)	150 L / 700–800 km	STUDY
Curb / design GVW	2,865 / 3,365 kg	5,150 / 6,150 kg	TARGET
GVW ceiling / margin	4,000 kg / 635 kg	7,000 kg / 850 kg	LOCKED / DERIVED
Packaging residual	–370 mm	–240 mm	OPEN — P0
Evidence level	L0	L0	BASELINE

2.1 Readiness assessment (hybrid host)

Dimension	Position	Principal exposure	Closure vehicle
Architecture	Defined and consistent at L0	No hardware or simulation evidence	SIM-001 (G1), tower bench (G2)
Range	Fuel-range screening with efficiency band (S7)	V-L 1,000 km holds only to ≈90–95 km/h average; η base 0.22	ENE-H-001, engine fuel maps
Mass	Closed at P50 with explicit reserve	Four-corner Class L P90 between ≈440 kg (RSS) and 548 kg (arithmetic) vs 380 kg budget	G1 tower-count decision, G2 supplier data
Packaging	Stack does not close	Height / cabin / ground-clearance re-baseline	G1 3-D datum stack
Safety	Invariants and screening FMEA	Fuel, heat and engine excitation near the cell	SAF-001, THERM-001 (G2)
Regulatory	ICE market exposure not assessed; V-L design GVW 3,365 kg crosses 3,500 kg threshold in some jurisdictions	Restrictions vary by region	REG-001 (G1)
IP	Ownership chain and filing OPEN	No filing receipt on record	Counsel, provisional filing

2.2 Revision history

Revision	Scope
R5.32	Combined module and hybrid-host baseline
R5.33	Hybrid host separated from module content; fuel-range screening; Class L sizing; V-H tower-count recommendation; ICE class TARGET
R5.34 (this issue)	All defects from the 2026-10-09 correction log closed: P90 mass wording (RSS vs arithmetic), six-tower mass penalty explicit, energy buffer for both 4- and 6-tower, packaging height ≥260/280 mm, fuel η band 0.18–0.30 (base 0.22) + aux load, range target cycle definition, lock/S4 logic clarification, document-control statement, Figure 1 caption, protection-package consistency, regulatory 3,500 kg note, status-tag definitions, high-severity RPN independence, IP wording.

2.3 Reference host comparison

The electric V-H design GVW equals the 7,000 kg ceiling, so payload is the balancing item; the hybrid hosts carry more margin. The V-H tower count differs (six in the electric baseline, four here as baseline with six-tower option); harmonisation is a G1 decision (REQ-H06).

Item	V-L electric	V-L hybrid	V-H electric	V-H hybrid
Powertrain	BEV, dual e-axle AWD	ICE V8 4.0 L TT + e-drive	BEV 6×6, three e-axes	ICE V12 6.0 L TT + e-drive
Peak power	≈350–380 kW	ICE ≈430–450 + e ≈150–180 kW	≈480–550 kW	ICE ≈460–500 + e ≈200–250 kW
On-board energy	216 kWh usable (baseline); 300 kWh option for ≥800 km	80 L + 20 kWh	300 kWh usable	150 L + 40 kWh

Item	V-L electric	V-L hybrid	V-H electric	V-H hybrid
Range target (STUDY)	≈900 km / 800 km with 300 kWh	1,000 km (≤90–95 km/h avg)	≈900 km utility	700–800 km
Towers	4 (Class S)	4 (Class S)	6 (Class L)	4 (Class L baseline); 6 option
Curb target	3,200 kg	2,865 kg	5,400 kg	5,150 kg
Design GVW	3,700 kg	3,365 kg	7,000 kg	6,150 kg
Margin to ceiling	300 kg	635 kg	0 kg	850 kg
Packaging residual	–370 mm	–370 mm	–240 mm	–240 mm
Dominant host risk	Pack energy density, HV safety	ICE range claim, fuel and heat	Pack mass; 900 km only at low speed	Four-corner P90 mass, fuel and heat
Evidence	L0	L0	L0	L0

3. Product definition and architecture

The host is defined around the CMFSB load and isolation architecture; it is not a powertrain retrofit. The interface between the two body pieces is the MCRS tower set — the sole mechanical connection in normal operation. Active isolation works inside a bounded mechanical safety envelope; comfort control, handling stiffness and crash containment are coordinated but independently bounded.

3.1 Layer stack (top to bottom)

Layer	Function	Safety role
Upper safety cell	Occupant and mission body (CFRP, roll structure, seats, closures)	Captured at all times
Micro stage (Lorentz)	High-bandwidth vibration isolation within ± 7.5 mm	Force-bounded; never a crash or horizontal path
Macro stage (hydraulic)	Leveling, ± 50 mm body-relative travel, lock seats	Maintains geometry and ride envelope
Progressive stops	Energy absorption on abnormal vertical events	Primary abnormal vertical energy path
SPORT positive lock	Pin, claw or hydraulic clamp at the seat band	Handling stiffness and load sharing; not a replacement for stops
Capture frame and links	Longitudinal, lateral and yaw restraint	Primary horizontal path
Lower chassis hardpoints	Load transfer into the host platform	Structural continuity

Figure 1 (conceptual) — Two-piece body: the upper cell is captured on MCRS towers above the lower host chassis. V-L: four towers (Class S) · V-H: four towers (Class L baseline); six-tower option under G1 decision; three towers shown for illustration. Vertical support is passive macro support; horizontal loads go through the capture frame and links; abnormal vertical energy through progressive stops.

3.2 Independent load and force paths

Load / function	Primary path	Secondary / containment path
Static vertical support	Passive hydro-pneumatic macro support	Progressive stops / chassis structure
COMFORT dynamic force	Lorentz micro actuator (force-bounded)	Macro stage + passive support
Large displacement and leveling	Hydraulic macro stage	Stops
SPORT stiffness	Positive locks + seat faces	Capture frame
Longitudinal / lateral / yaw	Capture frame, links, horizontal stops	SPORT lock faces
Crash / abnormal vertical	Progressive stops + structural hardpoints	Capture frame / links
Power loss	Passive hydraulic / compliant support (S4)	Mechanical capture
Battery crash load	Lower chassis + protected HV enclosure	Independent containment structures

The Lorentz stage is an actuator option. If replaced by another actuator family, the safety invariants and interface requirements remain applicable under configuration control.

3.3 Hold function

Hold is an optional, rate-limited secondary restraint (hydraulic orifice and/or friction device at the macro stage). It is distinct from the SPORT lock and from the stops, and is not a primary crash-energy path.

3.4 Qualitative comparison with conventional approaches

Approach	Comfort	Handling stiffness	Positive mechanical lock	Independent capture / crash path	Behaviour on power loss
Soft air / hydro-pneumatic	Good	Reduced in rapid manoeuvres	No	Limited to suspension travel and bump stops	Settles on bump stops
Adaptive dampers	Improved trade-off	Improved	No	No	Fixed or fail-soft damping
CMFSB / MCRS (target)	Targeted isolation (unproven at L0)	Targeted via SPORT lock	Yes (SPORT)	Yes — stops, capture frame, links	Settles on passive support, captured

Qualitative only; no measured superiority is claimed at evidence level L0.

4. Safety invariants and safety layers

Invariant	Rule
Mechanical capture	No operating state permits loss of geometric capture; no free levitation.
Static support	Vehicle weight on passive or fail-safe compliant paths — never the sole magnetic path.
Power loss (S4)	Passive support; powered leveling-supply and hold valves fail-closed; the hydro-pneumatic spring path (cylinder to accumulator) stays connected; software aliveness not required.
Crash path	Progressive stops + structure, independent of closed-loop magnetic force.
Mode transition	SPORT↔COMFORT never enters unconstrained free-float.
SPORT lock fault	Lock failure shall not defeat capture or progressive stops; SPORT is never declared with an incomplete lock set.
Horizontal restraint	Longitudinal / lateral / yaw loads resolved by dedicated paths and/or SPORT lock.
Override limit	Driver override cannot defeat the lateral-g interlock or a critical fault state.
Manual rescue	Two-step rescue available under power loss and hydraulic failure.
HV isolation	Traction HV domain galvanically isolated from the module control domain; a single HV fault shall not defeat capture or S4.
Thermal / fire	Battery runaway containment is a lower-chassis responsibility and shall not propagate into upper-body capture integrity.
Single-path rule	No single active-control path is credited as the primary crash path.
Lock state on power loss (clarified)	Locks remain engaged after power loss only if they were already engaged at the instant of loss (SPORT). From pure COMFORT the locks stay released or engage only at the seat band. Logic frozen in ICD-001 / LCK-001; added to REQ-M17 / REQ-M20.

4.1 Layered safety response

Safety layer	Failure addressed	Design response
Mechanical capture	Link / carriage failure	Capture frame with redundant restraint paths
Passive support	Power loss	Hydro-pneumatic / compliant support, fail-safe valves
Progressive stops	High-energy vertical event	Defined force–displacement curve (STR-001)
Horizontal restraint	Braking, cornering, crash	Dedicated links, stops and lock faces
Positive SPORT locks	Mode stiffness, lock integrity	Per-lock confirmation; incomplete set inhibits SPORT
HV safety	Electrical crash, isolation fault	HVIL, isolation monitor, service disconnect, crash inhibit
Thermal safety	Battery / inverter overheating	Redundant sensing, derating, propagation-resistant pack, vent routing
Control safety	Bad command, software fault	Bounds, diagnostics, degraded state, independent monitor

No ASIL assignment is claimed. E/E control improves ride and mode management but is not allowed to become the sole means of retaining the upper body. Items with severity ≥ 9 (loss of capture, link failure) require explicit pre-G2 review independent of RPN score.

5. Host definition and packaging gate

5.1 Geometry and mission

Parameter	V-L hybrid	V-H hybrid
Length / width (mirrors excl.)	5,000 / 2,000 mm	5,500 / 2,200 mm
Overall height (SPORT)	1,700 mm	2,000 mm
Wheelbase	3,000 mm	3,300 mm (front axle to mid of rear tandem)
SPORT ground clearance	200 mm	300 mm
Drive	4×4	6×6 (front + dual rear tandem)
Seats	4	6
Passenger cell	Full CFRP structural cell, budget ≤280 kg	Full CFRP structural cell, budget ≤420 kg
Roll structure	FIA-referenced CFRP / hybrid, ≤45 kg (no homologation claim)	Same, ≤70 kg
MCRS class / towers	Class S / 4	Class L / 4 (baseline); six-tower option under G1 (REQ-H06)
GVW engagement ceiling	4,000 kg	7,000 kg
Optional light protection package	—	≈200 kg; treated identically in both hosts for comparison; included in baseline curb or moved to reserve if omitted

Heights, ground clearance and cabin definition are engagement targets read as “FIXED-FOR-ENGAGEMENT / PACKAGING GATE OPEN”. Status tag definitions: BASELINE = frozen architecture decision; TARGET = design intent under evidence; STUDY = screening only; FIXED-FOR-ENGAGEMENT = locked for packaging studies but not packaging-closed; DERIVED = arithmetic consequence; OPEN-P0 = blocking packaging item.

V-H tower count (corrected analysis). The four-corner Class L P90 line-sum is 548 kg (arithmetic) against a 380 kg budget (+168 kg). RSS independent-corner estimate is ≈440 kg (+60 kg). The 150 kg reserve covers only the RSS case. Moving to six towers reduces per-tower load (≈109 → 73 kN at 20 g) but raises module mass from ≈380 kg to ≈565 kg (+≈185 kg), which also exceeds the 150 kg reserve. The six-tower recommendation is therefore justified only on load-per-tower grounds and must be evaluated together with the mass penalty and the DC-DC ≥4 kW requirement. Decision at G1 (REQ-H06).

5.2 Vertical packaging stack (P0 open item)

Element (mm)	V-L	V-H	Status
Overall height (SPORT)	1,700	2,000	FIXED-FOR-ENGAGEMENT
Ground clearance (SPORT)	200	300	FIXED-FOR-ENGAGEMENT
Available for structure (height – GC)	1,500	1,700	DERIVED
Chassis / axle structure	280	320	STUDY
Tower structure + mounts	120	140	STUDY
Macro travel (100) + free play (40)	140	140	STUDY
Floor-to-H-point	250	250	ASSUMPTION
Cabin, H-point to roof	1,000	1,000	TARGET
Roof / exterior shell	80	90	STUDY
Allocated sum	1,870	1,940	DERIVED
Residual margin	–370	–240	OPEN — P0

The module alone consumes ≥260 mm (V-L: tower 120 + travel 140) and ≥280 mm (V-H: tower 140 + travel 140) before stop stroke. FIXED-FOR-ENGAGEMENT geometry is an engagement target, not packaging closure. In external tables the heights read “FIXED-FOR-ENGAGEMENT / PACKAGING GATE OPEN”.

5.3 Illustrative closure path (to be validated at G1)

Lever (mm recovered)	V-L	V-H	Basis
Share tower depth with chassis / battery floor	60	60	Integration study
Reduce travel + free play allocation	30	30	Stop-stroke trade
Cabin H-point to roof 1,000 → 960	40	40	Ergonomic package
Floor-to-H-point 250 → 220	30	30	Seat / floor design
Shell thickness	10	10	CFRP roof design

Lever (mm recovered)	V-L	V-H	Basis
Total recovered	170	170	
Remaining residual	-200	-70	
Height needed at GC 200 / 300 mm	≈1,900	≈2,070	Re-baseline decision

Lever values are planning illustrations, not design commitments. Neither variant closes at the quoted height without a re-baseline of height, ground clearance or cabin definition. Stop stroke (STR-001) and micro-stage clearance are not yet in the stack and further reduce residual.

6. Powertrain and fuel system

6.1 Powertrain

Powertrain	V-L hybrid	V-H hybrid	Status
Architecture	Hybrid (ICE + e-drive)	Hybrid (ICE + e-drive)	Engagement target
ICE class	V8, 4.0 L twin-turbo class	V12, 6.0 L twin-turbo class	Engagement target
ICE power	≈430–450 kW class	≈460–500 kW class	TARGET
Electric traction	≈150–180 kW total	≈200–250 kW total	TARGET
Battery usable energy	20 kWh	40 kWh	Engagement target
Fuel tank	80 L	150 L	Engagement target
Combined range target	1,000 km (cycle avg ≤90–95 km/h)	700–800 km	STUDY

Cell chemistry, ICE supplier, transmission and e-drive suppliers are not frozen in Document 00; they belong to ICD-001 and supplier-controlled releases. Fuel-system, exhaust and heat-source locations are part of the G1 datum stack. The 48 V module domain is galvanically isolated from the host HV bus.

6.2 HV safety functions (hybrid HV battery and e-drive)

HV function	Baseline requirement
Battery enclosure	Protected, structural, serviceable, crash-contained; does not rely on the upper body for primary containment
HVIL	Continuous loop over pack, service disconnects, power electronics and charge interface
Isolation monitoring	Continuous during energized operation; fault response defined in ICD-001
Precharge	Controlled before inverter DC-link energisation
Crash isolation	Independent crash-signal path plus BMS / contactor logic
Service disconnect	Physically accessible and lockable
Thermal propagation	Cell-level detection, propagation-resistant pack design, vent routing
HV-to-MCRS separation	MCRS safe-state support remains available if traction HV is isolated
LV backup	Dedicated auxiliary energy sufficient for controlled safe-state functions

7. Fuel range screening

7.1 Assumptions (corrected)

Tank-to-wheel efficiency η base = 0.22 (lower bound 0.18, upper bound 0.30) until supplier fuel maps arrive (ENE-H-001). The previous 0.25–0.30 band was optimistic for a twin-turbo V8/V12 at very low load (≈ 15 kW from 430 kW, ≈ 3 % load). Auxiliary load (HVAC, hydraulics, thermal) of 1–2 kW is now included, consistent with the electric-host model.

Assumption	V-L	V-H	Note
Reference mass (curb + 200 kg)	3,065 kg	5,350 kg	Two occupants
CdA / Crr	0.86 m ² / 0.009	1.5 m ² / 0.012	As the electric host
Tank-to-wheel efficiency η	0.18–0.30 (base 0.22)	0.18–0.30 (base 0.22)	ASSUMPTION; hybrid assist not credited
Auxiliary load	1–2 kW	1–2 kW	HVAC + hydraulics + thermal
Fuel energy density	8.9 kWh/L	8.9 kWh/L	Gasoline-class, ASSUMPTION
Fuel volume	80 L	150 L	Battery credit excluded

Method: wheel energy per km = $(\frac{1}{2} \cdot \rho \cdot \text{CdA} \cdot v^2 + C_{rr} \cdot m \cdot g) \div 3.6$; fuel use (L/100 km) = (wheel energy + aux) $\div \eta \div 89$; range = tank $\div$ fuel use. Both η bounds are replaced by supplier brake-specific fuel maps in ENE-H-001.

7.2 Results (screening)

Case	Wheel Wh/km	L/100 km (η 0.30 / 0.22 / 0.18)	Range on fuel (η 0.30 / 0.22 / 0.18)
V-L 60 km/h	≈ 115	$\approx 4.3 / 5.9 / 7.2$	$\approx 1,860 / 1,360 / 1,110$ km
V-L 90 km/h	≈ 165	$\approx 6.2 / 8.5 / 10.3$	$\approx 1,300 / 940 / 780$ km
V-L 120 km/h	≈ 234	$\approx 8.8 / 12.0 / 14.6$	$\approx 910 / 670 / 550$ km
V-H 60 km/h	≈ 244	$\approx 9.2 / 12.5 / 15.3$	$\approx 1,640 / 1,200 / 980$ km
V-H 90 km/h	≈ 331	$\approx 12.4 / 17.0 / 20.7$	$\approx 1,210 / 880 / 720$ km
V-H 120 km/h	≈ 453	$\approx 17.0 / 23.2 / 28.4$	$\approx 880 / 650 / 530$ km

REQ-H01 (corrected). The 1,000 km target (V-L) and 700–800 km target (V-H) are defined for a specified cycle (average speed ≤ 90 –95 km/h for V-L). Monitoring at 90 and 120 km/h is for sensitivity reporting, not for confirming the target. V-L does not reach 1,000 km at 120 km/h under any η in the band.

8. Controlled mass budget

Curb mass is the arithmetic sum of the groups below plus an explicit programme reserve. Design GVW = curb + occupants + payload (payload derived). All figures are TARGET / STUDY at evidence level L0.

Group	V-L (kg)	V-H (kg)	Note
Upper body (CFRP cell + roll + seats + closures)	755	1,220	Incl. occupants policy mass not in curb
Battery (HV pack)	145	260	20 / 40 kWh usable class
Chassis, axles, wheels, brakes, steering	810	1,780	
ICE + hybrid drive + fuel + exhaust	580	820	≈20.2 % / 15.9 % of curb
MCRS module (4-tower Class L planning)	280	380	P50; see §9
HV/LV, cooling, protection	95	360	V-H includes optional 200 kg protection package
Reserve	70	150	≈2.4 % / 2.9 %
Curb total	2,865	5,150	TARGET
Occupants (policy)	400	600	100 kg each
Payload (derived)	100	400	
Design GVW	3,365	6,150	TARGET
Ceiling	4,000	7,000	LOCKED
Margin	635	850	DERIVED

Checks: V-L $2,795 + 70 = 2,865$; $2,865 + 400 + 100 = 3,365$. V-H $5,000 + 150 = 5,150$; $5,150 + 600 + 400 = 6,150$. The V-H budget includes the optional 200 kg protection package; if omitted it moves to reserve. Protection package is treated identically in the electric host for fair comparison (either both include or both exclude).

P90 four-corner mass (corrected). Arithmetic sum of Class L corner P90 values = $4 \times 137 = 548$ kg. RSS (independent corners) ≈ 440 kg. Excess versus 380 kg budget is therefore +60 kg (RSS) to +168 kg (arithmetic). The 150 kg reserve covers only the RSS case. Risk identifier: R-H2 (formerly referenced as R-07 in the R5.32 lineage).

Difference to electric curb (MCRS-FE baseline): Electric V-L ≈ +335 kg; Electric V-H ≈ +250 kg when protection package is treated consistently. Upper sprung mass incl. occupants and payload: V-L ≈ 1,255 kg; V-H ≈ 2,220 kg (without protection) or ≈ 2,420 kg (with protection).

Figure 4 (conceptual) — Curb-mass target by group. ICE + hybrid drive + fuel + exhaust share of curb: 20.2 % (V-L), 15.9 % (V-H). Labels must sum exactly to curb; HV/LV and reserve must appear. Axle description for V-L is 4×4, not three-axle 6×6.

9. Module mass, hydraulics and parameters

Module values are controlled by Master MCRS-DOC-00-R8.01 §7. The tables below are mirrored for standalone readability.

9.1 Corner assembly mass (Class L planning)

Element	P50 kg	P90 kg	Scope
Local structure (per tower)	18	25	Core
Lorentz + local carriage	22	30	Core
Sensors + local electronics	8	12	Core
Core tower subtotal	48	67	Target ≤55 kg P50 met by estimate; limit until G2 = 67 kg (conservative)
Cylinder + accumulator share	15	22	Corner
Valves / hoses / seals	5	8	Corner
Progressive stops	8	12	Corner
SPORT lock share	6	10	Corner
Harness / dampers / bellows	4	6	Corner
Pump + reservoir share (¼)	8	12	Shared (four-corner reference)
Full corner assembly (Class L)	94	137	STUDY; ≈95 kg in host WBS

Core-tower P90 limit: the previous ≈60 kg limit sat just above RSS (59.4 kg) and accepted independence. Conservative default until G2 supplier correlation data: 67 kg (arithmetic). The ≈60 kg limit is accepted only with correlation evidence.

9.2 Per-class corner mass and static support floor

Class	Mapped use	Corner P50	Four-corner P50	Static basis per corner	Floor (1.25 × Fmax)
S	Light host (four seats)	≈70 kg (target)	≈280 kg	Occupied share 310–350 kg	≥4.5 kN
M	Intermediate	≈85 kg	≈340 kg	Share 450–550 kg	≥7.0 kN
L	Heavy host	≈95 kg	≈380 kg	Reference max 787.5 kg	≥10.0 kN

10. Module electrical and auxiliary energy

Module supply is 48 V class, galvanically isolated from host HV. Peak RMS power for the micro-stage array exceeds the baseline DC-DC rating in both configurations:

- Four-tower: peak $\approx 6.3 \text{ kW} > 3 \text{ kW DC-DC}$ $\rightarrow$ local energy buffer (48 V battery or supercapacitor) is mandatory.
- Six-tower: peak $\approx 9.45 \text{ kW} > 4 \text{ kW DC-DC}$ $\rightarrow$ local energy buffer is mandatory.

Therefore a local energy buffer is required for both four-tower and six-tower hosts. Energy reserve for safe-state functions is verified under ENE-H-001 (hybrid) / ENE-E-001 (electric). Power-loss detection and valve closure timing are frozen in ICD-001.

10.1 Electrical domain partition

Domain	Voltage	Isolation	Function
Module control / Lorentz	48 V class	Galvanic from host HV	Micro actuators, sensors, BCE
Hydraulic valves / hold	48 V / 12 V	Fail-closed on loss	Leveling, hold, lock actuation
Host traction HV	High voltage	Isolated from module	e-drive, HV battery
Host LV	12 V	Standard vehicle	Lighting, body, ICE control
Local energy buffer	48 V	Module domain	Peak shaving for Lorentz; mandatory both configs

10.2 Auxiliary energy budget (screening)

Auxiliary load for range screening includes HVAC, hydraulic pump (when active), thermal management and control electronics. Baseline screening uses 1–2 kW continuous. Peak Lorentz demand is covered by the local buffer, not by continuous DC-DC rating. Buffer energy and recharge rate are sized in ENE-H-001 so that a worst-case isolation event sequence does not deplete the buffer below the safe-state reserve.

On power loss the module shall retain sufficient passive hydraulic energy and mechanical capture without drawing on the local buffer or host LV. The buffer is for comfort performance, not for the S4 safety case.

11. Control system, modes and state machine

11.1 Controller partition

Domain	Responsibility
Vehicle domain controller	Driver intent, drive state, vehicle dynamics, high-level requests; cannot bypass MCRS interlocks
MCRS Body Control ECU (BCE)	Mode arbitration, corner coordination, command bounds, lock sequencing, leveling, diagnostics
Independent safety monitor	Force / stroke / mode bounds, critical interlocks, watchdog and plausibility, where adopted
Corner controllers	Local sensing and actuator current / valve commands, bounded by BCE
BMS	Battery state, HV safety, contactor control, thermal limits
Brake controller	Primary service braking and regenerative coordination

Comfort control may be sophisticated; safety bounds remain simple, deterministic and independently monitorable. The monitor may be called “independent” only after power, clock, processor, software, communication and sensor dependencies are analysed and tested. OTA or calibration changes shall not bypass configuration control.

11.2 State machine

State	Name	Rule
S0	Safe / captured / power-off	No powered release; passive capture retained (equivalent to S4)
S1	Precharge / self-test	Capture healthy; auxiliary power path checked
S2	COMFORT	Macro + micro isolation active; capture mandatory
S2S	SPORT	Seat band; all required positive locks confirmed; stages held
S3	Degraded	Reduced authority; force and travel bounded
S4	Power-loss capture	Passive support; fail-safe valves; no software dependency for retention
S5	Impact preparation	High-rate unlock inhibited; stops prepared
S6	Impact / arrest	Progressive stops active; no actuator authority required
S7	Post-event safe	No automatic re-enable; manual clearance required
S8	Maintenance lockout	Physical lockout; stationary only

Any powered state → S4 on power loss (engaged locks remain engaged only if already locked; from COMFORT locks stay released). S4 exits only to S1 after power restore and self-test. S2↔S2S only under the lateral-g policy with complete lock confirmation. Full transition table including S5–S8 exit conditions (manual clearance) is required in ICD-001. Detection of power loss ($t_{det,max}$) must be hardware/passive so that “S4 without software” is not contradicted.

12. Power-loss behaviour and lateral-g policy

12.1 REQ-M07 power-loss structure

After loss of commanded active power the system shall: (1) detect the loss within $t_{det,max}$ (hardware/passive path); (2) enter S4; (3) close the relevant valves within $t_{valve,max}$; (4) remain inside the capture envelope; (5) stay below z_{max} and v_{max} ; (6) prevent uncontrolled free-float; (7) settle in the defined S4 hydraulic state with passive support at or above the class static floor, with residual capacity and environment stated, under electrical loss and the defined hydraulic failure cases. State definition alone is not acceptance.

12.2 Lateral-g policy (REQ-M21)

Element	Rule
Inhibit	SPORT→COMFORT inhibited immediately when $ ay > 0.35$ g, for automatic and manual requests
Re-enable	Only after $ ay $ stays below 0.30 g for the hold time
Driver override	May queue a request; cannot defeat the interlock
Filter and latency	Validated filter and measurement-to-inhibit latency budget, both biased to inhibit
Sensor faults	Validity, plausibility, disagreement thresholds, stale-data handling; a single lateral sensor cannot alone trigger an unsafe transition

*Thresholds are temporary TARGET values; SIM-001 and HIL-001 freeze final thresholds, hysteresis, filtering and latency.
PARAM-M21-ROLL-GRAD to be defined at G2.*

13. Structural principles

Structural screening loads are based on upper sprung mass, not GVW. Current corner loads assume equal share ($\div 4$ or $\div 6$). This is a magnitude screen only. STR-001 must calculate real corner distribution including CG location, occupant/payload placement and load transfer under 1 g braking or cornering.

Progressive stops provide the primary abnormal vertical energy path with a defined force–displacement curve (STR-001). Capture frame and links provide the primary horizontal path. SPORT locks are not a substitute for stops.

13.1 Screening load cases (upper sprung mass)

Case	V-L upper mass	V-H upper mass	Per-corner (equal share)	Note
1 g static	≈1,255 kg	≈2,220 kg (w/o prot.)	≈3.1 / 5.4 kN	Occupied + payload
2 g vertical	same	same	≈6.2 / 10.9 kN	Screening
20 g crash vertical	same	same	≈62 / 109 kN (4-tower)	Stops path
20 g (6-tower)	—	same	≈73 kN	Load reduction basis for 6-tower
1 g braking transfer	TBD	TBD	TBD	STR-001 — not equal share
Cornering transfer	TBD	TBD	TBD	STR-001 — CG dependent

The 20 g four-tower per-corner load of ≈109 kN on V-H is the primary quantitative driver for the six-tower option. With six towers the same case falls to ≈73 kN. Mass penalty of the sixth pair of towers (≈185 kg module) must be weighed against this load reduction at G1 (REQ-H06).

13.2 Progressive stop envelope

Stops shall absorb the design abnormal vertical energy without reliance on the Lorentz actuator or closed-loop force. Force–displacement curve, residual height and rebound are frozen in STR-001. Stop stroke is not yet in the packaging stack and further reduces residual margin; it is an OPEN item for G1.

14. Requirements (selected hybrid-host)

Full requirements register lives in the Master and ICD-001. Selected hybrid-host items:

ID	Statement (summary)	Status
REQ-H01	Range target 1,000 km (V-L) / 700–800 km (V-H) defined for cycle average ≤ 90 –95 km/h (V-L); 90/120 km/h are sensitivity only	OPEN — ENE-H-001
REQ-H06	V-H tower count: four (baseline) or six; decision at G1 with both load-per-tower and mass-penalty criteria	OPEN — G1
REQ-M07	Power-loss structure (detection, valves, envelope, settle)	OPEN — ICD-001
REQ-M17/M20	Lock state on power loss: engage only if already locked; COMFORT stays released	OPEN — LCK-001
REQ-M21	Lateral-g interlock 0.35 / 0.30 g	TARGET (temporary)
REG-001	Regulatory exposure including V-L design GVW 3,365 kg crossing 3,500 kg threshold in some markets	OPEN — G1

15. Safety analysis

Screening FMEA exists at L0. No HARA/FTA or independence argument yet. Closure vehicle: SAF-001 (G2). High-severity ($S \geq 9$) items (loss of capture, link failure) require explicit pre-G2 review independent of RPN. RPN scoring does not hide high-severity rows.

15.1 Screening FMEA summary (selected rows)

ID	Failure mode	Effect	S	O	D	RPN	Action
F-01	Loss of geometric capture	Upper body free	10	2	4	80	Pre-G2 review; redundant paths
F-02	Link / carriage fracture	Horizontal path lost	9	2	5	90	Pre-G2 review; STR-001
F-03	SPORT lock incomplete set	False SPORT declaration	8	3	3	72	Per-lock confirmation logic
F-04	Power-loss valve fails open	Loss of passive support	9	2	4	72	Fail-closed design; test
F-05	HV fault propagates to module	Capture defeated	9	2	5	90	Galvanic isolation; HVIL
F-06	Fuel / heat near cell	Thermal event	8	3	4	96	THERM-001; packaging
F-07	Single lateral sensor fault	Unsafe mode transition	7	3	4	84	Plausibility + dual path
F-08	Software command out of bounds	Force exceedance	7	3	3	63	Independent monitor

Severity ≥ 9 rows are flagged for mandatory pre-G2 independent review regardless of RPN. Full FMEA is in SAF-001.

16. Verification

Evidence ladder L0–L5. Current position L0 (architecture). Gate sequence G0–G5. SIM-001 (G1) and tower bench (G2) are the first hardware/simulation vehicles. All transitions of the state machine require HIL evidence before credit.

16.1 Evidence ladder

Level	Meaning	Typical artefacts
L0	Architecture definition, arithmetic consistency	This baseline, mass/energy tables, invariants
L1	First-order simulation / screening	SIM-001, road-load model, packaging stack
L2	Component / subsystem bench	Tower bench, actuator characterisation, hydraulic failure
L3	HIL / system integration	State-machine HIL, power-loss timing, interlocks
L4	Vehicle-level prototype	Instrumented mule, range test, structural test
L5	Production-intent / certification	Homologation evidence, ASIL if claimed

16.2 Gate sequence (summary)

Gate	Focus	Key exit criteria
G0	Document release	Owner release; reviewer appointed
G1	Architecture freeze / packaging	3-D datum stack; tower-count decision; ICD-001 draft; REG-001
G2	Subsystem evidence	Tower bench; SAF-001; supplier data; ACT/STR/ENE closure
G3	System HIL	Full state machine; power-loss; lateral-g
G4	Vehicle mule	Instrumented range and structural tests
G5	Production readiness	Beyond this baseline

17. Gates, risks and open items

17.1 Open items (selected)

#	Item	Closure
1	Packaging residual –370 / –240 mm (P0)	G1 3-D datum stack; re-baseline height/GC/cabin
2	V-H tower count (4 vs 6) with mass and load criteria	G1 REQ-H06
3	P90 module mass (RSS vs arithmetic); supplier correlation	G2 supplier data
4	Fuel maps and η confirmation	ENE-H-001
5	Lock / S4 logic from COMFORT	ICD-001 / LCK-001
6	Full state-machine transition table S5–S8	ICD-001
7	Regulatory 3,500 kg threshold for V-L	REG-001 G1
8	IP filing receipt	Counsel
9	Independent technical reviewer appointment	≤2026-12-05 or G1
10	Energy buffer sizing (4- and 6-tower)	ENE-H-001 / ICD-001
11	Stop stroke and micro-stage clearance in packaging stack	STR-001 / G1
12	Structural corner load distribution (CG, transfer)	STR-001
13	High-severity ($S \geq 9$) pre-G2 review	SAF-001
14	PARAM-M21-ROLL-GRAD definition	G2
15	Protection package inclusion policy (both hosts)	G1 mass closure

Risk R-22 (packaging P0): residual is certainly negative; probability set to 5 so that $L \times I \geq 20$. Risk R-H2: four-corner P90 mass excess (formerly R-07 in R5.32 lineage).

17.2 Risk register (selected)

ID	Risk	L	I	Score	Mitigation
R-22	Packaging residual remains negative	5	4	20	G1 re-baseline; levers in §5.3
R-H2	Four-corner P90 mass exceeds reserve	4	4	16	G1 tower decision; G2 supplier
R-H3	Fuel range claim fails at higher speed	4	3	12	ENE-H-001; cycle definition
R-H4	ICE heat / exhaust near cell	3	4	12	THERM-001; packaging
R-H5	Regulatory 3,500 kg threshold	3	3	9	REG-001; market selection
R-H6	Lock logic ambiguity on power loss	3	4	12	ICD-001 / LCK-001
R-H7	No filing receipt (IP)	3	3	9	Counsel; provisional
R-H8	Energy buffer undersized	3	3	9	ENE-H-001 sizing

18. Controlled document tree

Master MCRS-DOC-00-R8.01 (module values) → Hybrid MCRS-DOC-00-R5.34 (this file) and Electric MCRS-FE-DOC-00-R7.xx. ICD-001, SIM-001, ENE-H-001, SAF-001, STR-001, LCK-001, REG-001, THERM-001 are the primary work packages. This document is the controlling hybrid-host Document 00 under the Master; it does not control module values.

Document / WP	Role	Owner / gate
MCRS-DOC-00-R8.01	Master — module values	Owner; governs module
MCRS-DOC-00-R5.34	Hybrid host baseline (this)	Owner; hybrid values
MCRS-FE-DOC-00-R7.xx	Electric host baseline	Owner; electric values
ICD-001	Interface control document	G1 draft
SIM-001	System simulation	G1
ENE-H-001	Hybrid energy / fuel maps	G1/G2
SAF-001	Safety analysis / HARA	G2
STR-001	Structural / stops	G2
LCK-001	Lock logic and confirmation	G1/G2
REG-001	Regulatory exposure	G1
THERM-001	Thermal / exhaust packaging	G2
ACT-001	Actuator characterisation	G2

19. IP, disclosure and commercial paths

Founder / Legal Owner: Amirbahador Atai (individual rights holder). Assignment to a legal entity is a separate controlled act. Licensable module: MCRS-MOD-001. Paths A/B/C as defined in §1.2. Wording: integration into a two-piece vehicle body (not “EV body”). Trade-mark reference: MCRS (not “MCRS FULL ELECTRIC”). No filing receipt on record; provisional filing under counsel is OPEN.

19.1 Commercial paths (detail)

Path A (primary): paid Option covering SIM-001 run and ICD-001 draft. Buyer may bring its own host. No vehicle sale is implied. Module licence terms are under counsel.

Path B: host engagement studies using the hybrid (or electric) reference host geometry, mass and packaging. Deliverables are interface and packaging studies; no vehicle sale.

Path C: IP-transaction discussion under counsel only (T3 classification). No technical disclosure outside NDA.

19.2 Disclosure control

This document is T1 Confidential / NDA controlled. Distribution is limited to parties under valid NDA. Public disclosure of module architecture or claims beyond the claims-discipline table is prohibited until filing and owner release.

20. Claims discipline and glossary

20.1 Claims discipline

Claim type	Allowed statement	Forbidden statement
Range	"≈1,000 km target under defined cycle (STUDY)"	"1,000 km range", "verified range"
Safety	"Invariants defined; screening FMEA at L0"	"ASIL-rated", "crash-certified"
Comfort	"Targeted isolation (unproven at L0)"	"Superior ride", "clinical outcome"
Mass	"P50 closed with reserve; P90 OPEN"	"Mass closed", "production mass"
Packaging	"FIXED-FOR-ENGAGEMENT / OPEN-P0"	"Locked geometry", "packaging closed"
Readiness	"TRL 2–3, evidence L0"	"Production ready", "homologated"

20.2 Glossary (selected)

Term	Definition
CMFSB	Captured Mechanical Floating Safety Body — the isolated upper safety cell architecture
MCRS	Modular Captured Ride System — the product family and licensable module
MCRS-MOD-001	Licensable four- or six-tower isolation module
COMFORT (S2)	Isolation mode: locks released, macro + micro active
SPORT (S2S)	Locked mode: all required positive locks confirmed, stages held
S4	Power-loss capture: passive support, fail-safe valves, no software dependency for retention
Class S / L	Module corner mass / load class (light / heavy host)
P50 / P90	50th / 90th percentile mass estimate
RSS	Root-sum-square combination of independent corner uncertainties
FIXED-FOR-ENGAGEMENT	Locked for packaging studies but not packaging-closed
OPEN-P0	Blocking packaging open item
L0–L5	Evidence ladder levels (architecture → certification)
G0–G5	Programme gates
RA-1	Reference application: ambulance / specialty medical transport (comfort context only; no clinical claim)

End of MCRS-DOC-00-R5.34 Hybrid Host Engineering Baseline (Corrected). All defects listed in the 2026-10-09 correction log that apply to the hybrid host have been closed or explicitly carried as OPEN with closure vehicle.

This issue supersedes R5.33 for all hybrid-host content. Module values remain governed by Master MCRS-DOC-00-R8.01.

Document control: T1 NDA Controlled · Evidence level L0 · TRL 2–3 · NOT CERTIFIED · Page count ≥24

Appendix A — Correction log summary (R5.33 → R5.34)

This appendix records every defect from the 2026-10-09 correction list that applies to the hybrid host, together with the action taken in R5.34. Electric-only items are noted as deferred to the electric baseline revision.

#	Defect (summary)	Action in R5.34
2	Six-tower recommendation ignored mass penalty	Explicit +185 kg mass and load criteria; G1 REQ-H06
3	P90 four-corner = 548 kg only	RSS ≈440 / arithmetic 548; reserve covers RSS only
4	Energy buffer only for six-tower	Mandatory for both 4- and 6-tower
5	Packaging module height ≥260 only	≥260 mm V-L, ≥280 mm V-H
9	Lock vs S4 from COMFORT contradiction	Clarified; REQ-M17/M20; ICD/LCK
10	Document control statement	Hybrid under Master; module values by Master
11	Figure 1 caption six towers	Four baseline; six option; caption corrected
12	One value one place	Host numbers only here; module by citation
13	ID consistency	MCRS-MOD-001 uniform
15	Protection package unfair compare	Identical treatment both hosts
18	Risk R-07 reference	Now R-H2
19	Fuel η optimistic	Base 0.22; band 0.18–0.30; aux 1–2 kW
20	REQ-H01 range at 120 km/h	Target for ≤90–95 km/h cycle only
23	Core tower P90 limit 60 kg	Conservative 67 kg until G2
24	Equal-share structural loads	Screen only; STR-001 for real distribution
26	Figure 4 labels	Percentages and axle description corrected
28	Status tags undefined	Defined in §5.1
29	R-22 probability low	L=5; score ≥20
30	High S independent of RPN	Pre-G2 review mandated
31	State machine S5–S8 incomplete	OPEN; ICD-001 full table
33	3,500 kg regulatory note missing	Added in §2.1 and REG-001
34	IP / trademark wording	two-piece vehicle body; MCRS

Electric-only defects (battery P90, GVW circular definition, peak power split, AC charger times, pack density, WLTP label, FE-IDs, etc.) are deferred to the next electric-host baseline revision and are not closed in this hybrid document.

Appendix B — Status tag definitions (normative)

Tag	Meaning	Change control
BASELINE	Architecture decision frozen for this issue	Requires revision and owner release
TARGET	Design intent under evidence; may move with data	G1/G2 evidence can refine
STUDY	Screening value only; not design commitment	Superseded by supplier / test data
FIXED-FOR-ENGAGEMENT	Locked for packaging / engagement studies; not packaging-closed	Re-baseline at G1 if stack fails
DERIVED	Arithmetic consequence of other controlled values	Recalculated when inputs change
OPEN — P0	Blocking packaging open item	Must close or re-baseline before G1 exit
ASSUMPTION	Explicit assumption with owner and closure route	Closed by named work package
LOCKED	Ceiling or hard constraint (e.g. GVW ceiling)	Programme policy change required

Previous use of the word “LOCKED” for open packaging heights was misleading and has been replaced by FIXED-FOR-ENGAGEMENT throughout this issue.

— End of Appendices —

MCRS-DOC-00-R5.34 · Hybrid Host Engineering Baseline (Corrected) · T1 NDA Controlled · Not Certified